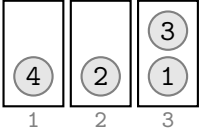
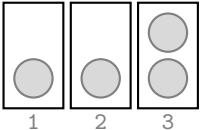
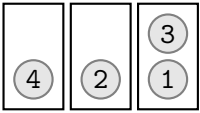
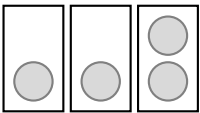


	n balls	m bins	any placement allowed	no more than one ball per bin $m \geq n$	at least one ball in every bin (use all bins)
	dist	dist	m^n	$m(m-1) \cdots (m-n+1)$	$S(n, m) m!$
	indist	dist	$\binom{n+m-1}{n}$	$\binom{m}{n}$	$\binom{n-1}{m-1}$
			weak composition	m choose n	strong composition
	dist	indist	$S(n, m) + \cdots + S(n, 1)$	1	$S(n, m)$
	indist	indist	$p_{\leq m}(n)$	1	$p_m(n)$

Adapted from Federico Ardila, *Enumerative Combinatorics*, Lecture 10.

$S(n, m)$ is the number of set partitions of $[n]$ into m blocks,
that is, Stirling numbers of the second kind.

$p_m(n)$ counts integer partitions of n with exactly m parts.

$p_{\leq m}(n)$ counts integer partitions of n with no more than m parts.

The binomial coefficient is $\binom{a}{b} = \frac{a!}{b!(a-b)!}$